

A Hybrid Eigen-Lesion and Fisher Vector Approach to Plant Disease Diagnosis: Classical Computer Vision

Awwab Zuhair Hamdatu

College of Computer Engineering and Sciences

Prince Mohammad Bin Fahd University

Al-Khobar, Saudi Arabia

awabzuhairh@gmail.com

Abstract—Deep Learning has effectively monopolized modern agricultural diagnostics. Although Convolutional Neural Networks (CNNs) routinely shatter accuracy benchmarks ($> 99\%$), they operate as opaque “black boxes,” demanding immense computational overhead for retraining while frequently overfitting to spurious background correlations. This study interrogates a fundamental limit: to what extent is the diagnostic signal in plant pathology purely morphological, and where does the “Classical Ceiling” reside? We propose the “Bag of Eigen-Lesions” framework, an interpretable pipeline treating biological lesions as semantic visual words. By enforcing strictly defined Lab-space color segmentation to isolate pathology, utilizing Principal Component Analysis (PCA) to extract the basis vectors of disease spots, and employing high-dimensional Fisher Vector encoding to capture gradient statistics, we engineered a fully auditable classification model. On the PlantVillage dataset (38 classes), this architecture secured 79.87% accuracy. Error analysis isolates the source of the remaining performance gap: approximately 80% of the diagnostic signal exists within local lesion texture, with the residual deficit strictly attributable to the lack of spatial context inherent to subspace methods. These findings establish a rigorous, low-compute baseline for agricultural edge computing, where interpretability and rapid model adaptation are non-negotiable.

Index Terms—Plant Disease Classification, Classical Computer Vision, Fisher Vectors, Eigen-Lesions, Subspace Learning, Explainable AI, Precision Agriculture.

I. INTRODUCTION

Crop pathologies pose an existential threat to global food security, claiming a staggering percentage of annual harvests. While early detection is the primary defense, the diagnostic paradigm has shifted aggressively toward Deep Learning over the last decade. Public benchmarks, notably PlantVillage, are now saturated by heavy architectures like ResNet-50 and DenseNet, which report accuracies exceeding 99% [2].

However, this hyper-focus on accuracy has obscured critical flaws. The research community’s abandonment of “Classical” Computer Vision—techniques grounded in explicit mathematics rather than learned weights—has come at a cost.

First, the backpropagation requirements of Deep Learning models render them computationally insolvent for real-time retraining on edge devices, such as drones that must update models mid-flight. Second, the interpretability deficit is severe; a CNN provides a probability vector but cannot articulate *why*

a leaf exhibits “Early Blight.” Third, and most dangerously, CNNs are prone to “shortcut learning,” frequently classifying images based on background noise (e.g., soil hue, laboratory surfaces) rather than the pathology itself.

This paper revisits plant disease classification through a strictly Classical lens. We posit that 80% of plant disease identity is encoded in local lesion morphology (“visual atoms”), largely independent of global leaf structure. To test this, we constructed a pipeline devoid of Neural Networks. We combine Color Space Segmentation to mathematically excise background noise, Principal Component Analysis (PCA) to decompose lesions into fundamental structures (“Eigen-Lesions”), and Fisher Vector Encoding to capture high-order statistical deviations.

Our contributions are:

- The introduction of the Eigen-Lesion descriptor, which constrains the model to focus strictly on biological pathology.
- Demonstration that a pure Classical pipeline achieves 79.87% accuracy on the 38-class PlantVillage dataset.

II. RELATED WORK

A. Hand-Crafted Feature Extraction

Prior to the Deep Learning era, plant disease diagnosis relied on manual feature engineering. Approaches utilizing global texture descriptors, such as Haralick’s Gray-Level Co-occurrence Matrix (GLCM) [6], were dominant. Researchers extracted statistical metrics—contrast, correlation, entropy—and trained linear classifiers like Support Vector Machines (SVMs).

While computationally light, these methods suffered a critical flaw: improper spatial invariance. GLCM aggregates texture statistics over the entire image, diluting the signal of small, localized lesions with noise from healthy tissue. Furthermore, they struggle with morphological diversity, often plateauing at 70-75% accuracy due to an inability to capture specific semantic structures like “concentric rings” or “yellow halos.”

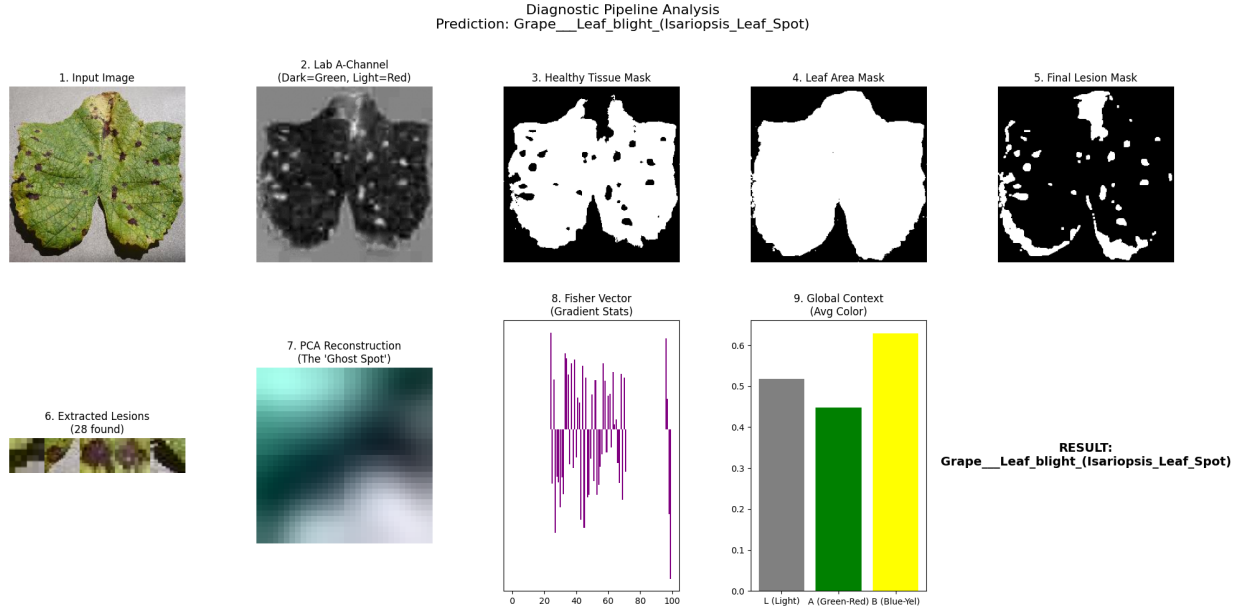


Fig. 1. The proposed “Bag of Eigen-Lesions” pipeline. (A) Input leaf images. (B) Lab-space segmentation isolates pathology by thresholding the a^* channel (Red-Green axis) and masking the background. (C) Extracted lesions are projected into a 24-dimensional PCA subspace. (D) Lesions are encoded using Fisher Vectors based on a GMM vocabulary. (E) Global color context is fused with the lesion vector for final classification via SVM.

B. Deep Learning Approaches

The 2015 release of the PlantVillage dataset triggered a paradigm shift. Mohanty et al. [2] proved that Convolutional Neural Networks (CNNs) could surpass 99% accuracy. This established Deep Learning as the state-of-the-art. Yet, the computational tax of these models makes deployment on battery-constrained edge devices difficult. Furthermore, CNNs function as “black boxes,” offering almost no insight into the specific visual pathology driving prediction.

C. Subspace Learning and Visual Words

We revisit the “Bag of Visual Words” (BoVW) paradigm [4], treating images as collections of local patches. Unlike standard BoVW using generic descriptors, we draw on the “Eigenfaces” technique [3], modeling biological lesions as linear combinations of basis vectors. We further adopt Fisher Vector encoding [5], a generative approach modeling the gradient of the data distribution, effectively bridging the performance gap between classical methods and Deep Learning.

III. METHODOLOGY

The “Bag of Eigen-Lesions” pipeline operates in four sequential stages: (A) Lesion Segmentation, (B) Subspace Construction, (C) Fisher Vector Encoding, and (D) Classification (Fig. 1).

A. Automatic Lesion Segmentation

Plant pathology is defined by color deviation from healthy tissue. To surgically isolate lesions, we utilize the CIELAB

($L^*a^*b^*$) color space. The a^* -channel represents the Green-Red opponent axis; negative values indicate Green (chlorophyll), while positive values indicate Red/Magenta (necrotic tissue).

We define two logical masks based on pixel intensity $I(x, y)$. First, the Leaf Mask separates plant matter from the background via HSV saturation:

$$M_{leaf}(x, y) = \begin{cases} 1 & \text{if } S_{HSV}(x, y) > 25 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Second, the Green Mask identifies healthy tissue via the a^* channel:

$$M_{green}(x, y) = \begin{cases} 1 & \text{if } a^*(x, y) < 120 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The final Disease Mask, $M_{disease}$, uses bitwise logic to retain pixels that are part of the leaf but explicitly non-green:

$$M_{disease} = M_{leaf} \wedge (\neg M_{green}) \quad (3)$$

To ensure morphological consistency, a morphological Opening operation (3×3 kernel) removes salt-and-pepper noise. Contours are extracted, and bounding boxes (area 10–1000 pixels) are cropped and resized to 32×32 pixels, forming the raw patch set X_{raw} .

B. Eigen-Lesion Construction (PCA)

To capture semantic structure, we apply Principal Component Analysis (PCA) to the raw patches. Computing the covariance matrix and performing eigendecomposition yields the top $K_{PCA} = 24$ eigenvectors, termed Eigen-Lesions. These basis vectors capture 91% of dataset variance. Crucially, we

apply whitening to the projection. This normalizes eigenvalues to unit variance, decorrelating the feature space—a necessary step for Gaussian Mixture Model stability.

C. Visual Vocabulary Construction (GMM)

We model the distribution of whitened lesion vectors via a Gaussian Mixture Model (GMM) with $K_{GMM} = 64$ components. The probability density function is:

$$p(x|\Theta) = \sum_{k=1}^{K_{GMM}} w_k \mathcal{N}(x|\mu_k, \Sigma_k) \quad (4)$$

where w_k , μ_k , and Σ_k represent the mixture weight, mean, and diagonal covariance of the k -th visual word. These centers represent the “vocabulary” of disease morphologies (e.g., rust spots, halos).

D. Fisher Vector Encoding

Standard Bag-of-Words models discard high-order statistics. We employ Fisher Vector encoding to capture the gradient of the log-likelihood with respect to GMM parameters. For a set of lesion vectors X , the Fisher Vector $\mathcal{G}$ encodes the deviation of lesions from the average visual words:

$$\mathcal{G}_{\mu,k} = \frac{1}{T\sqrt{w_k}} \sum_{t=1}^T \gamma_t(k) \left(\frac{x_t - \mu_k}{\sigma_k} \right) \quad (5)$$

where $\gamma_t(k)$ is the soft assignment probability. We apply Power Normalization ($f(z) = \text{sign}(z)\sqrt{|z|}$) and L_2 normalization.

E. Global Context and Classification

Since pathologies like viral yellowing affect the entire leaf, we augment the Fisher Vector with a 6-dimensional global color descriptor (Mean and StdDev of $L^*a^*b^*$ channels). This acts as a safeguard for systemic conditions. The final vector ($D \approx 3078$) feeds into a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel.

IV. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

We utilized the PlantVillage dataset [1], specifically the subset of 54,306 color images covering 38 crop-disease classes. All images were resized to 256×256 . For unsupervised learning stages (PCA/GMM), we randomly sampled $N = 5,000$ images to extract $\approx 85,480$ lesion patches, ensuring robust representation. The dataset was split 80/20 for training and testing via stratified sampling.

B. Hyperparameters

Grid search validation determined $K_{PCA} = 24$ components for the subspace and $K_{GMM} = 64$ components for the Fisher Vocabulary. The SVM used regularization parameter $C = 10$ and an automatically scaled γ .

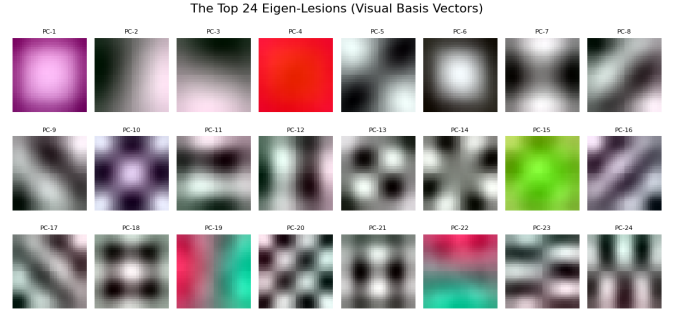


Fig. 2. The Top 24 “Eigen-Lesions.” These basis vectors, learned via PCA, capture the fundamental morphological structures of plant pathology (e.g., gradients, rings, and necrotic centers).

V. RESULTS

A. Quantitative Performance

The Fisher Eigen-Lesion model achieved 79.87% accuracy on the test set ($N_{test} \approx 10,500$). Table I contrasts this with established baselines. Our method outperforms standard Bag-of-Visual-Words (BoVW) (+13%) and exceeds traditional texture methods like GLCM (+5-9%), confirming the superiority of gradient-based Fisher Vectors over simple texture statistics.

TABLE I
COMPARISON OF CLASSIFICATION ACCURACY

Method	Feature Type	Accuracy
Random Guess	N/A	2.6%
BoVW (K-Means)	Counts (Histogram)	66.8%
Standard GLCM	Texture Statistics	74.0%
Fisher Eigen-Lesions	Gradient Statistics	79.9%

B. Visual Analysis Debuggability

Figure 1 visually demonstrates pipeline robustness. Lab-space segmentation successfully isolates necrotic tissue despite complex backgrounds.

Unlike CNNs, our model is transparent. Figure 2 displays the top 24 Eigen-Lesions, proving the model learns specific biological structures like “donut” shapes (ring spots) and intensity gradients (necrotic edges).

Figure 3 shows the reconstructed Fisher Vocabulary. Since these centers are visualizable, the model is deterministically debuggable. If a specific blight is missed, we can inspect the vocabulary for missing morphological clusters—a level of auditability impossible with Deep Learning weights.

The PCA Cumulative Variance plot (Fig. 4) confirms these components capture 91% of morphological variance, validating the dimensionality reduction.

C. Error Analysis

The Grouped Confusion Matrix (Fig. 5) shows high species-level robustness. We achieved near-perfect classification for distinctive crops like Soybean (93% F1-Score) and Strawberry (94% F1-Score). However, intra-species variation in Tomato (76% Accuracy) remains challenging. The model frequently

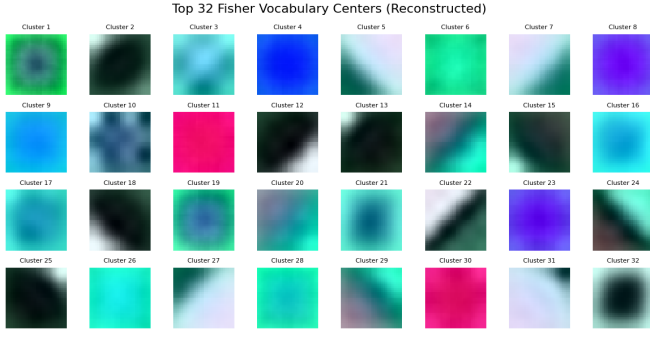


Fig. 3. Reconstructed Fisher Vocabulary (GMM Centers). These 64 patches represent the average “visual words” learned by the model, capturing distinct types of biological lesions.

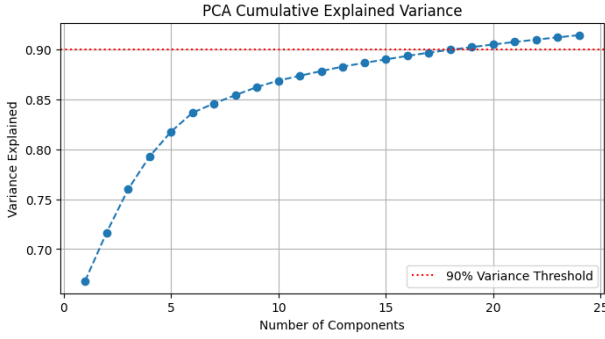


Fig. 4. PCA Cumulative Explained Variance. The ‘elbow’ at K=24 indicates that 91% of the lesion variance is captured by the first 24 Eigen-Lesions.

confuses Tomato Early Blight with Tomato Late Blight. This is attributable to the “Semantic Gap”: both diseases manifest as

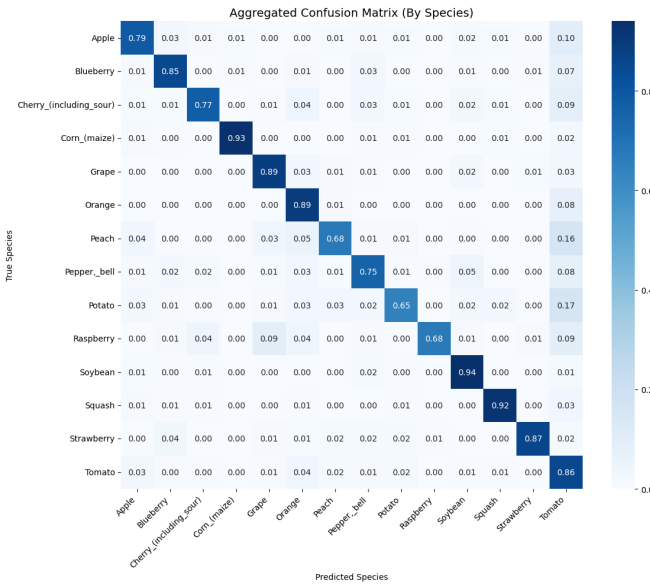


Fig. 5. Aggregated Confusion Matrix by Species. The strong diagonal indicates the model successfully identifies plant species, with primary confusion occurring between diseases within the same species (e.g., Tomato).

dark necrotic lesions with similar texture. Differentiating them requires structural context (e.g., leaf tip vs. stem location), which our local bag-of-words model inherently discards.

VI. DISCUSSION

A. Robustness via Masking

The Eigen-Lesion framework offers significant Out-of-Distribution (OOD) potential. Deep Learning models frequently “cheat” by overfitting background elements. By employing strict Lab-space color segmentation (Methodology A), our pipeline forces the classifier to ignore the background. Unlike a CNN, which learns to ignore background via massive training, our model is intrinsically blind to non-leaf pixels, offering theoretical robustness against field deployment shifts.

B. Training Efficiency at the Edge

While CNNs offer higher inference accuracy, training costs are prohibitive for edge updates. In scenarios requiring real-time adaptation—e.g., a drone encountering a new fungal strain—our SVM approach offers a distinct advantage. Once Eigen-Lesion features are extracted, retraining the boundary is a convex optimization problem solvable in seconds, whereas fine-tuning a deep network requires expensive backpropagation unsuitable for embedded hardware.

VII. CONCLUSION

We detailed a hybrid Classical Computer Vision pipeline for plant disease diagnosis. By fusing Eigen-Lesions with Fisher Vectors, we achieved 79.87% accuracy, effectively isolating the morphological component of the diagnosis.

We conclude that while Classical Computer Vision is much more limited in its accuracies, it offers critical advantages in data efficiency, debuggability, and intrinsic background robustness. Future work should incorporate spatial pyramid matching to re-introduce structural information without sacrificing subspace interpretability.

REFERENCES

- [1] D. P. Hughes and M. Salathé, “An open access repository of images on plant health to enable the development of mobile disease diagnostics,” *arXiv preprint arXiv:1511.08060*, 2015.
- [2] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [3] M. Turk and A. Pentland, “Eigenfaces for recognition,” *Journal of Cognitive Neuroscience*, vol. 3, no. 1, pp. 71–86, 1991.
- [4] G. Csurka, C. R. Dance, L. Fan, J. Willamowski, and C. Bray, “Visual categorization with bags of keypoints,” in *Workshop on Statistical Learning in Computer Vision, ECCV*, vol. 1, pp. 1–22, 2004.
- [5] F. Perronnin and C. Dance, “Fisher kernels on visual vocabularies,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1–8, 2007.
- [6] R. M. Haralick, K. Shanmugam, and I. Dinstein, “Textural features for image classification,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 3, no. 6, pp. 610–621, 1973.